

4.2.2. Bratu problem. We use lrAA to solve the non-linear Bratu problem

$$u_{xx} + u_{yy} + \lambda e^u = 0, \quad (x, y) \in [0, 1] \times [0, 1],$$

with $\lambda = 1$ and homogeneous Dirichlet boundary conditions. To find the approximate solution we discretize this equation using standard second order finite difference approximations for the x and y derivatives. Given an approximation $X(i, j) \approx u(x_i, y_j)$ this results in a function $G_B(i, j; X)$ describing the equation

$$(4.2) \quad \begin{aligned} G_B(i, j; X) &= \frac{1}{h_x^2} (X(i+1, j) - 2X(i, j) + X(i-1, j)) \\ &+ \frac{1}{h_y^2} (X(i, j+1) - 2X(i, j) + X(i, j-1)) + \lambda e^{X(i, j)}. \end{aligned}$$

Near the boundaries some of the terms in this expression will be set to zero to account for the homogeneous Dirichlet boundary conditions. In the numerical examples below we take the mesh to be $x_i = ih_x$, $h_x = \frac{1}{m+1}$ and $y_j = jh_y$, $h_y = \frac{1}{n+1}$ with $m = n = 200$, making the setup the same as in [52].

The fixed point function $G(i, j)$ is obtained by applying the preconditioned Richardson iteration. We have

$$X^{k+1}(i, j) = G(i, j; X^k, \alpha) \equiv X^k(i, j) + \alpha M(G_B(i, j; X^k)).$$

We test lrAA with no preconditioner and with the ES preconditioner (corresponding to `Rel1_x_n10.1E10`) described above. The lrAA parameters used are the following $\text{TOL} = 10^{-6}$, $\hat{m} = 5$, $\theta = 0.9$, $\alpha = 0.125h_x^2$ (un-preconditioned case) and $\alpha = 0.1$ (preconditioned case, no scheduling is used). In all experiments we use a rank 1 matrix with Frobenius norm around 1.

The results in Figure 11 display the numerical solutions obtained by lrAA methods with and without the ES preconditioner. In particular, both methods obtain visually similar numerical results in terms of solution contours and column and row index section for the final iterates. The un-preconditioned lrAA gives a monotonically increasing intermediate ranks. The ES-preconditioned lrAA converges very rapidly in 8 iterations.

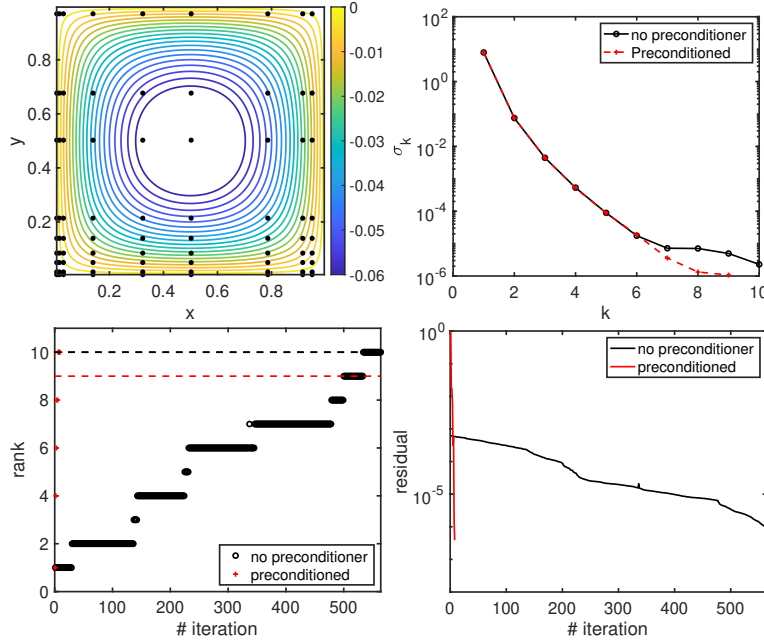


FIG. 11. Bratu problem solved by lrAA. The top left figure displays the contour levels of the converged solution along with markers at the intersection points of the final index sets $\mathcal{I}$ and $\mathcal{J}$ (for both lrAA with and without ES preconditioner). The top right figure displays the singular values from lrAA solutions with and without the ES preconditioner. The bottom left figure and right figures display the rank evolution and the decay of the residual throughout the lrAA iterations for lrAA with and without the ES preconditioner.